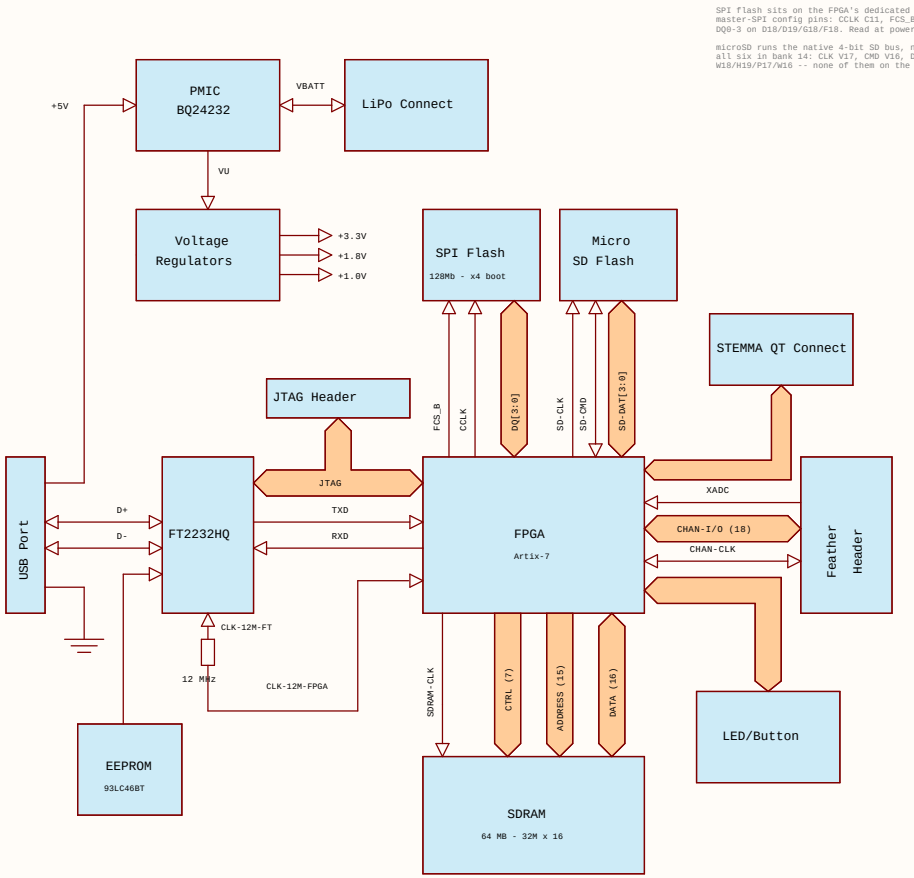
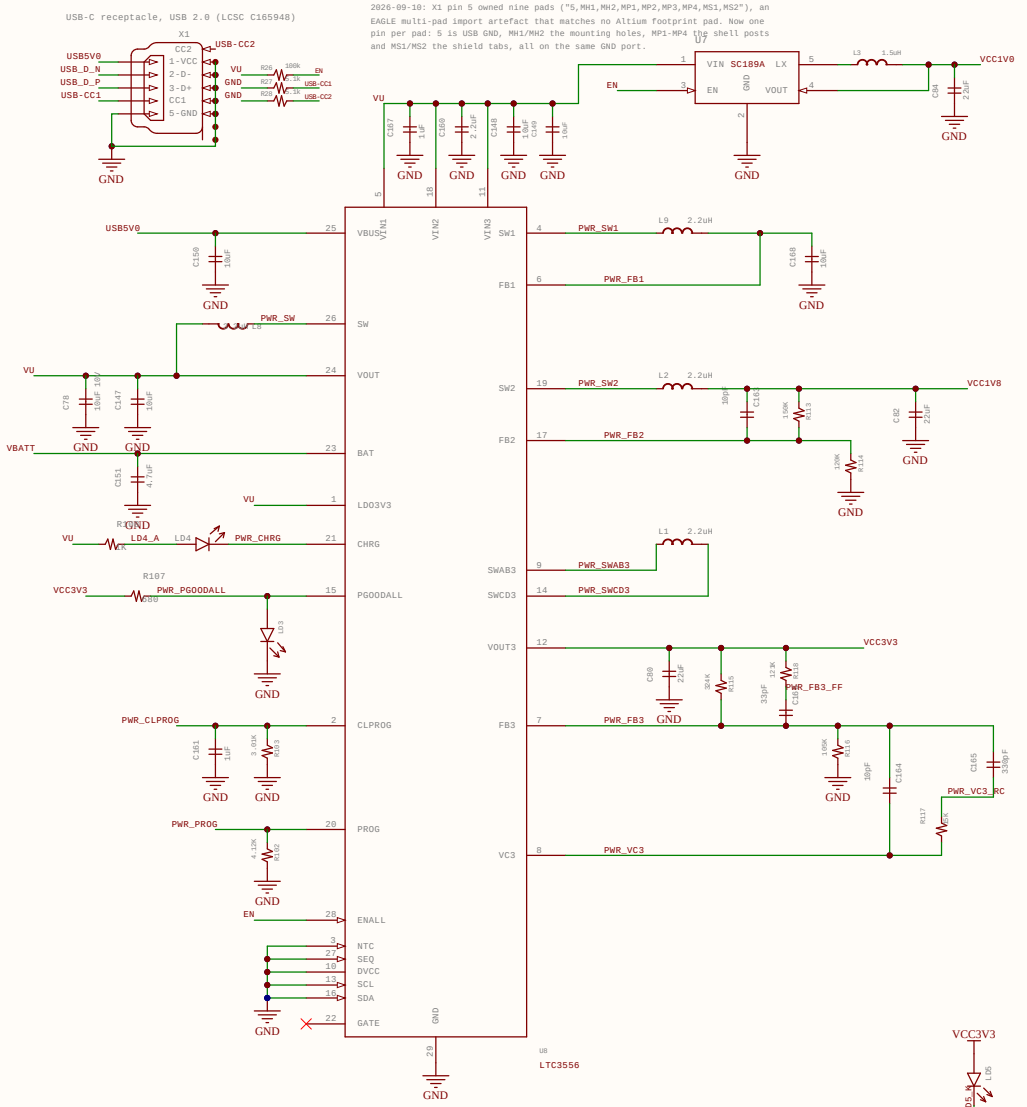
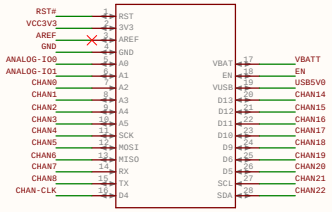


Block Diagram			
PROJECT:	asar_a7	REV:	9/2/2026 1:29:21 PM
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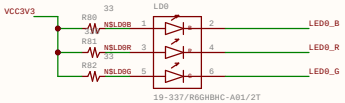

USB-C receptacle, USB 2.0 (LCSC C165948)
2026-09-10: X1 pin 5 owned nine pads ("5,MH1,MH2,MP1,MP2,MP3,MP4,MS1,MS2"), an EAGLE multi-pad import artefact that matches no Altium footprint pad. Now one pin per pad: 5 is USB GND, MH1/MH2 the mounting holes, MP1-MP4 the shell posts and MS1/MS2 the shield tabs, all on the same GND port.
U7
SC189A
VIN
LX
EN
GND
VOUT
VCC1V0
U8
LTC3556
VBUS
SW
VOUT
BAT
LDO3V3
CHRG
PGOODALL
CLPROG
PROG
ENALL
NTC
SEQ
DVCC
SCL
SDA
GATE
SW1
FB1
SW2
FB2
SWAB3
SWCD3
VOUT3
FB3
VC3
VCC3V3
PWR_SW1
PWR_FB1
PWR_SW2
PWR_FB2
PWR_SWAB3
PWR_SWCD3
PWR_FB3_FF
PWR_VC3_RC
PWR_VC3
PWR_SW
PWR_CHRG
PWR_PGOODALL
PWR_CLPROG
PWR_PROG
USB5V0
VBATT
USB-CC1
USB-CC2
JST PH 2-pin, 1S LiPo
2026-11-10: U8 bq24232 + U5 TPS63802 + U6 SC189L -> one Analog Devices LTC3556 (USB PowerPath, charger 248 mA, Buck2 VCC1V8 400 mA, buck-boost VCC3V3 1 A); VCC1V0 stays on U7 SC189A (1.5 A; XPE puts a half-full XC7A35T at ~0.4 A). U7 starts on ENALL; Buck1 runs unloaded as a sequencing slot (SEQ = GND), so 1V0 -> 1V8 -> 3V3. I2C off (DVCC/SCL/SDA = GND); USB input limit = 1x mode set by R103 (3.01k placeholder, value TBD). NTC grounded, GATE open. LD3 = rails good, LD4 = charging.
Power Supplies
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X2: Feather headers, a 1x16 and a 1x12 strip, 0.1 in pitch, rows 0.8 in (20.32 mm) apart.
Pads 1-16 form the 16-pin strip, 17-28 the 12-pin strip; nets keep the Zulu A7 CHANn names.
16-PIN: 1 RST (RST=: PROGRAM_B V10 and the reset button), 2 3V3 (VCC3V3), 3 AREF (no load),
  4 GND, 5 A0 and 6 A1 (ANALOG-IO0/1 -> the XADC dividers on AD4 and AD12),
  7-10 A2-A5 = CHAN0-CHAN3, 11 SCK = CHAN4, 12 MOSI = CHAN5, 13 MISO = CHAN6, 14 RX = CHAN7,
  15 TX = CHAN8, 16 D4 = CHAN-CLK (ball P18, clock-capable).
12-PIN: 17 VBAT (VBATT, the LiPo on X4), 18 EN (EN pins of U5-U7), 19 VUSB (USB5V0 from X1),
  20-24 D13, D12, D11, D10, D9 = CHAN14-CHAN18, 25 D6 = CHAN19, 26 D5 = CHAN20,
  27 SCL = CHAN21, 28 SDA = CHAN22.
FPGA balls: vivado/asar_feather_pins.xdc and the pin table in README.md.

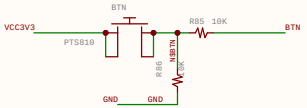
Feather headers: 1x16 + 1x12, 0.1 in



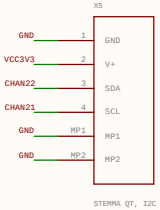
2026-09-10: BTN pads 1-2 are one switch terminal and 3-4 the other. They were one pin each ("1,2" and "3,4"), an EAGLE multi-pad import artefact that matches no Altium footprint pad; now four pins, each pad on its partner's net.



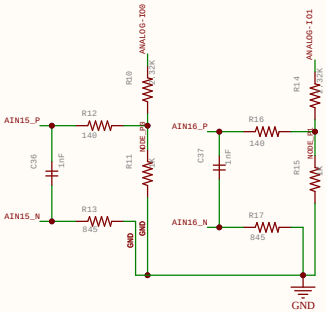








STEMMA QT, I2C

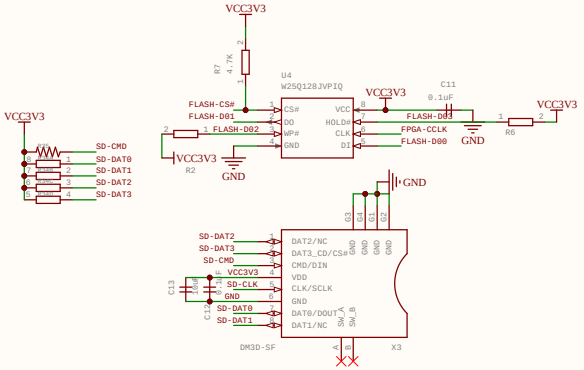


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General IO			
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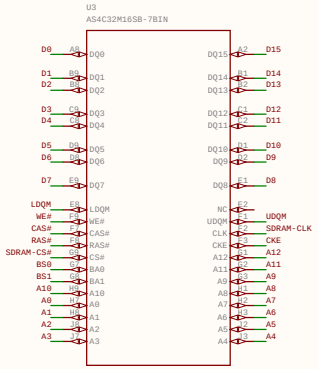
MASTER SPI BOOT (x4). M2:M1:M0 = 0:0:1 -- R20 pulls M0 to VCC3V3, R21/R22 pull M1/M2 to GND -- so at power-up the FPGA drives CCLK itself and clocks its bitstream out of U4 with no USB and no host present. That mode can only use the dedicated configuration pins: CCLK C11, FCS_B K19, D00_MOSI D18, D01_DIN D19, D02 G18, D03 F18. No other ball will do, which is why SDRAM A4 had to move off K19 (it is now on R3, bank 34). After DONE the six pins revert to ordinary user I/O. R2/R6 keep WP#/HOLD# high through the x1 command phase and through the float window before the FPGA drives D02/D03 (PUDC_B is tied high, so the FPGA has no internal pull-ups). R7 keeps FCS_B deselected. Vivado: set_property CONFIG_MODE SPIx4 [current_design] set_property BITSTREAM.CONFIG.SPI_BUSWIDTH 4 [current_design] The microSD is deliberately NOT on this bus any more: an inserted card sits in native SD mode watching bitstream traffic go past on what it believes is CMD, and one 48-bit window with a valid CRC7 is enough to make it answer and drive the line the FPGA is reading. X3 has its own CLK/CMD/DAT0 on V17/V16/W18 (bank 14), clear of the six master-SPI pins.

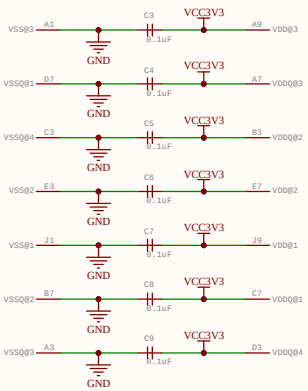


microSD runs the NATIVE 4-BIT SD BUS, not SPI. CLK, CMD and DAT0-DAT3 -- six signals, all in bank 14: CLK V17, CMD V16, DAT0 W18, DAT1 H19, DAT2 P17, DAT3 W16. DAT3 is the old SPI chip-select, which needs no new pin. R34/R35 are the 4.7k pull-ups the SD spec requires on CMD and DAT0-DAT3. R9 (100R on DAT0 only) was removed: one damped line out of four is a skew, not termination.

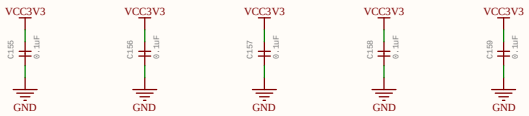
2026-09-06: X3 is now the Hirose DM3D-SF push-pull socket (HRS 609-0025-8), DM3 catalog p9. Same 8-contact pinout; shell pads G1-G4 to GND; card-detect switch A/B (closed with card) left open. Needs its own footprint on the PCB.

U3 = AS4C32M16SB-7BIN: 512Mbit (32M x 16) = 64MB, 54-ball FBGA, 8 x 8 mm, 0.8 mm ball pitch. The column address is A0-A9 (1024 columns): the memory controller must be set to 10 column bits, or it reaches only half the array. Row A0-A12, banks BA0/BA1, refresh 8192 cycles / 64 ms. -7BIN: 143 MHz max at CL3, tRCD/tRP 21, tRC 63 ns, TA -40 to 85 C (industrial), IDD1 max 110 mA.

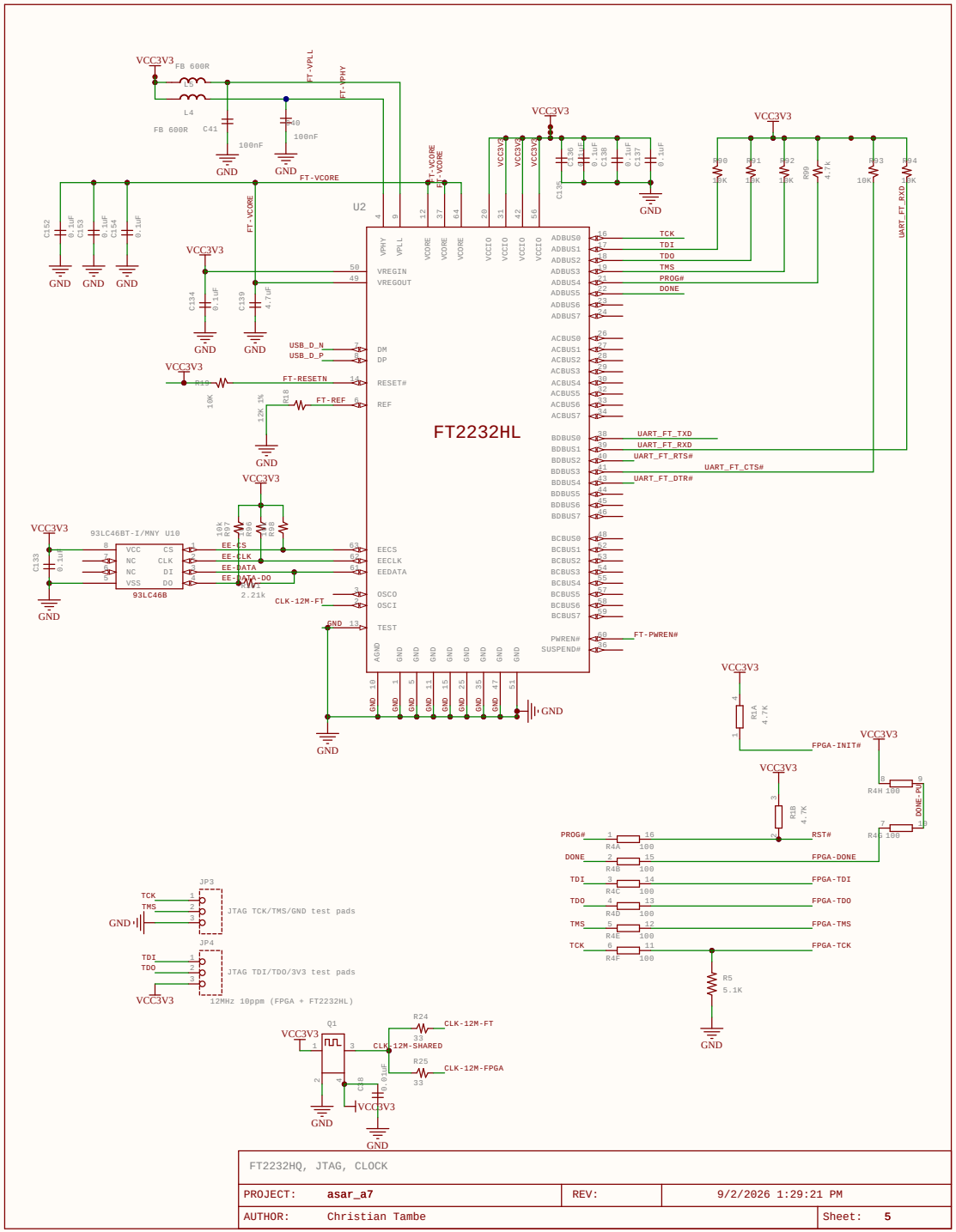


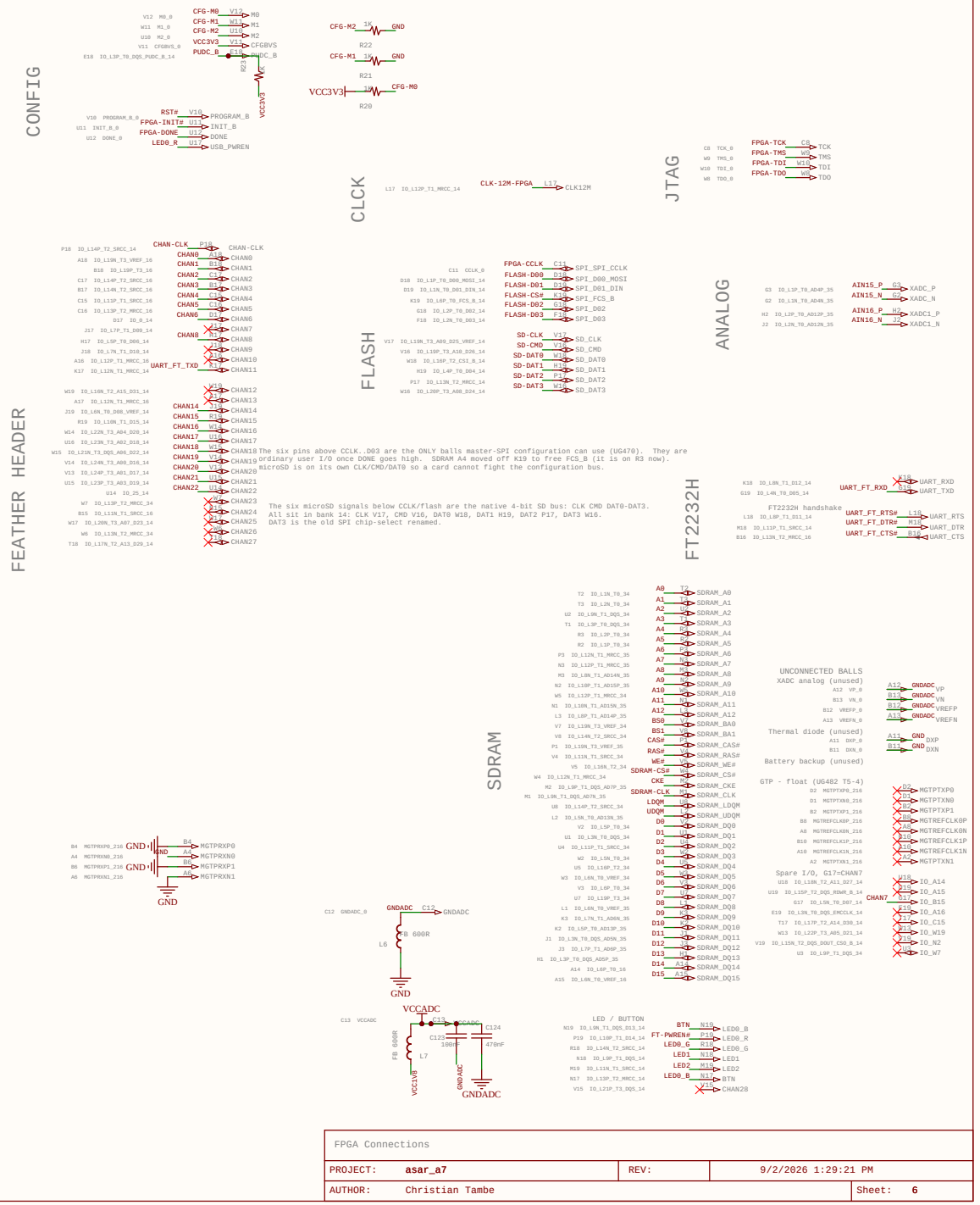


U3 DECOUPLING -- U3 is the 54-ball FBGA. Each supply ball has a ground of its own domain at the mirror position in its row: VDD A9/E7/J9 to VSS A1/E3/J1, VDDQ A7/B3/C7/D3 to VSSQ A3/B7/C3/D7. Alliance isolates VDDQ from VDD for DQ noise immunity; on this board both are VCC3V3 and both grounds are GND. C3-C9, C11 and C12 (0.1 uF) and C13 (10 uF bulk) are on the bottom side under U3. Do NOT split the ground plane.



Memory			
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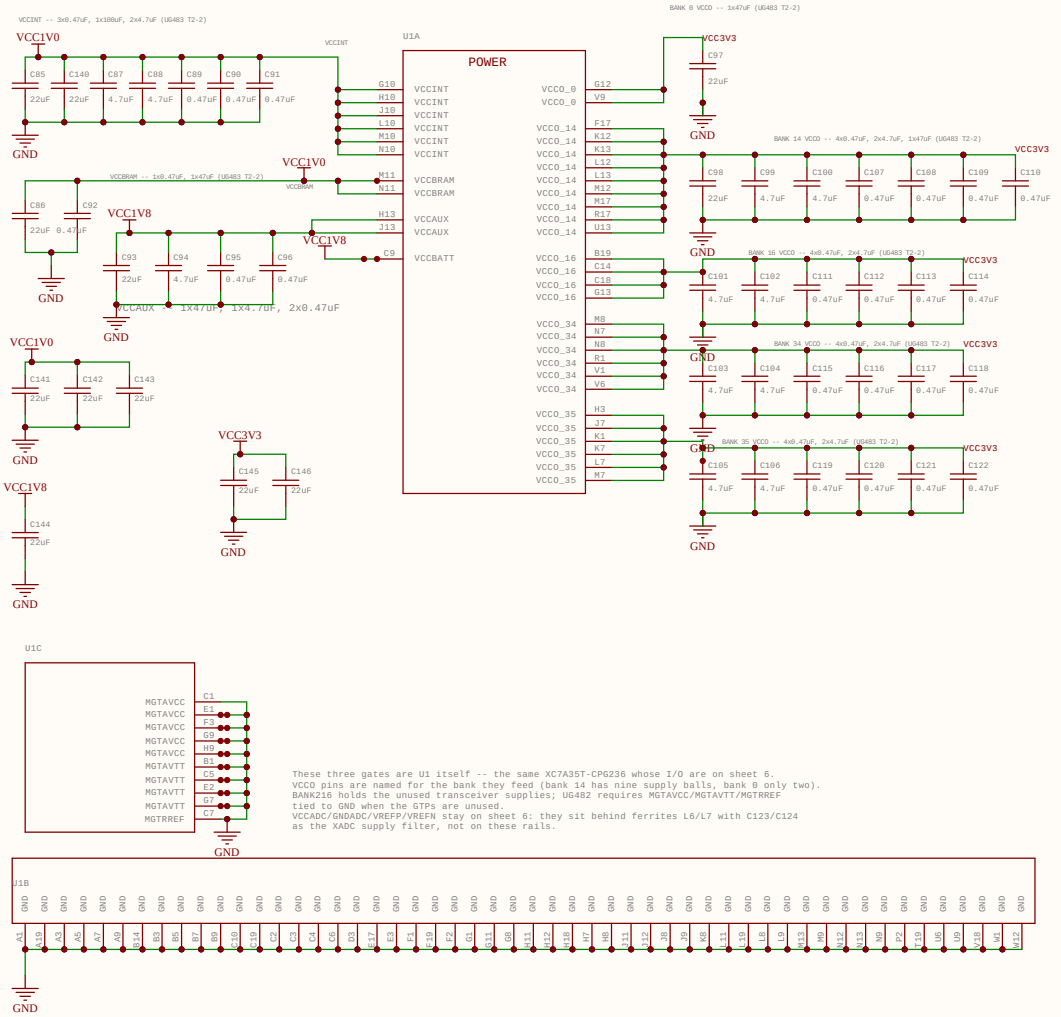

FT2232HL
U2
93LC46B
JTAG TCK/TMS/GND test pads
JTAG TDI/TDO/3V3 test pads
12MHz 10ppm (FPGA + FT2232HL)
FT2232HQ, JTAG, CLOCK
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The six pins above CCLK..D03 are the ONLY balls master-SPI configuration can use (UG470). They are ordinary user I/O once DONE goes high. SDRAM A4 moved off K19 to free FCS_B (it is on R3 now). MicroSD is on its own CLK/CMD/DAT0 so a card cannot fight the configuration bus.

The six microSD signals below CCLK/flash are the native 4-bit SD bus: CLK CMD DAT0-DAT3. All sit in bank 14: CLK V17, CMD V16, DAT0 W18, DAT1 H19, DAT2 P17, DAT3 W16. DAT3 is the old SPI chip-select renamed.

FPGA Connections			
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FPGA POWER			
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